

Reform Project

import the data and use janitor to clean names

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(sandwich)
library(tseries)
```

Registered S3 method overwritten by 'quantmod':

```
method      from
as.zoo.data.frame zoo
```

```
library(tidyverse)
library(lubridate)
library(strucchange)
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

```
as.Date, as.Date.numeric
```

Attaching package: 'strucchange'

The following object is masked from 'package:stringr':

```
boundary
```

```
library(stargazer)
```

Please cite as:

Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
R package version 5.2.3. <https://CRAN.R-project.org/package=stargazer>

```
library(lavaan)
```

This is lavaan 0.7-2

lavaan is FREE software! Please report any bugs.

```
library(readr)
library(modelsummary)
# Load time_series_GB_20040101_0000_20260422_0807 data
time_series_GB_20040101_0000_20260422_0807 <- read_csv("c:/Users/User/Downloads/time_series_GB_20040101_0000_20260422_0807.csv")
```

Rows: 268 Columns: 9

— Column specification —

Delimiter: ","

dbl (8): Immigration, Reform UK, Labour Party, Conservative Party, Scottish...

date (1): Time

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
data_clean <- time_series_GB_20040101_0000_20260422_0807 %>%
  mutate(Time = ymd(Time))
data_clean = janitor::clean_names(data_clean)
data_clean$reform_uk = time_series_GB_20040101_0000_20260422_0807$`Reform UK`+time_series_GB_20040101_0000_20260422_0807$`UKIP`
#Combine Reform and UKIP topics since reform wasn't founded until later on to create a composite variable
```

do a preliminary structural break check on the immigration time series, we are going to be using pre defined dates though

```
summary(strucchange::breakpoints(data_clean$immigration~1))
```

Optimal (m+1)-segment partition:

Call:

```
breakpoints.formula(formula = data_clean$immigration ~ 1)
```

Breakpoints at observation number:

m = 1	57		
m = 2	57	228	
m = 3	57	162	228
m = 4	40	80	162 228
m = 5	40	80	120 162 228

Corresponding to breakdates:

```

m = 1  0.212686567164179
m = 2  0.212686567164179
m = 3  0.212686567164179 0.604477611940298
m = 4  0.149253731343284 0.298507462686567 0.604477611940298
m = 5  0.149253731343284 0.298507462686567 0.447761194029851 0.604477611940298

```

```

m = 1
m = 2  0.850746268656716
m = 3  0.850746268656716
m = 4  0.850746268656716
m = 5  0.850746268656716

```

Fit:

```

m    0    1    2    3    4    5
RSS 8475 6026 3562 3365 3270 3180
BIC 1697 1617 1487 1483 1487 1491

```

define break dummies using known immigration related topics such as brexit referendum

```

data_clean <- data_clean %>%
  mutate(breakdBp1 = ifelse(time > as.Date("2019-04-01"), 1, 0), # Reform UK founded small boar
         breakpukip= ifelse(time>as.Date("2015-05-01"),1,0),      #Brexit #
         breakpukipl= ifelse(time>as.Date("2012-01-01"),1,0)) # Hostile Enviroment Policy#

```

pre define immigration and inflation break coefs

```

data_clean = data_clean %>% mutate(Refbreakim = immigration*breakdBp1,
Brexitbreakim = immigration*breakpukip,
Hostilebreakim = immigration*breakpukipl)
data_clean = data_clean %>% mutate(Refbreakinf = inflation*breakdBp1,
Brexitbreakinf = inflation*breakpukip,
Hostilebreakinf = inflation*breakpukipl)

```

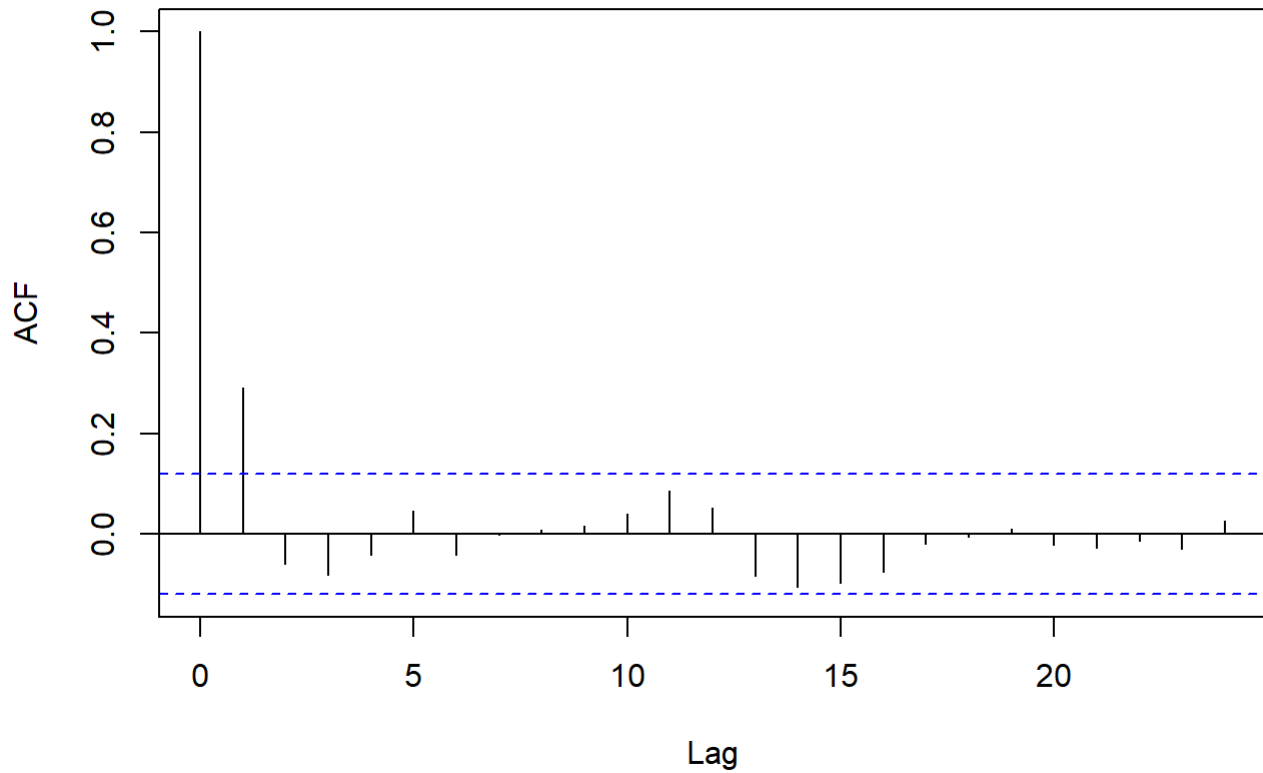
define simple independent ols models and check autocorrelations of residuals

```

library(sandwich)
library(lmtest)
breakmodellim = lm(reform_uk~breakdBp1+breakpukip+breakpukipl+immigration+Refbreakim+Brexitbre
acf(breakmodellim$residuals)

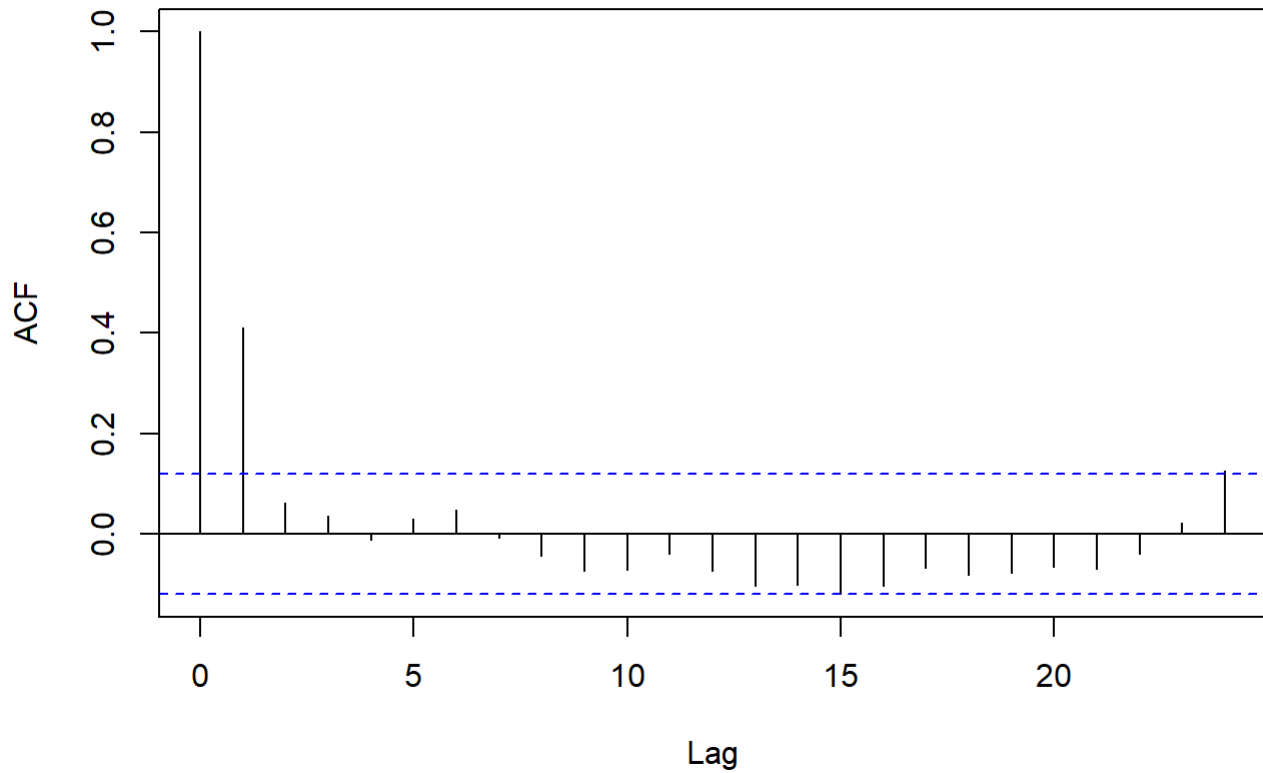
```

Series breakmodel1im\$residuals



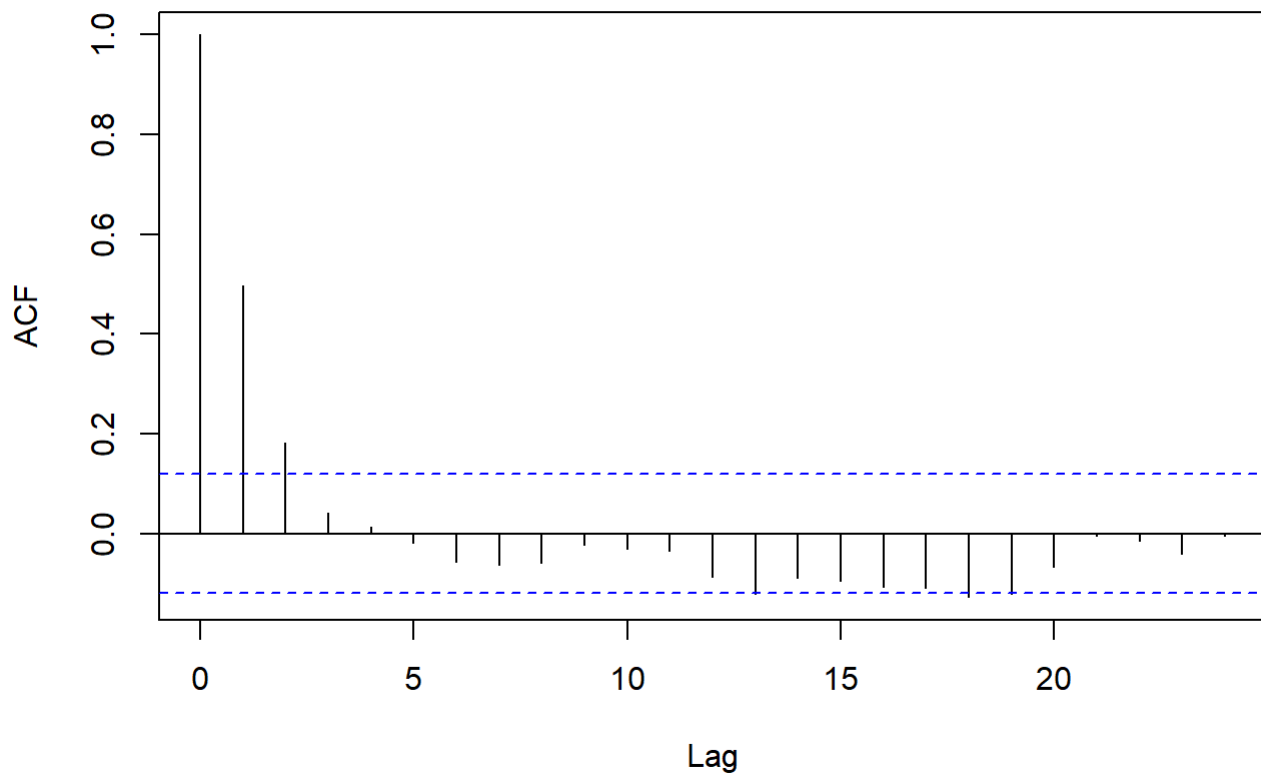
```
breakmodel2im = lm(conservative_party~breakdBpl+breakpukip+breakpukipl+immigration+Refbreakim+  
acf(breakmodel2im$residuals)
```

Series breakmodel2im\$residuals



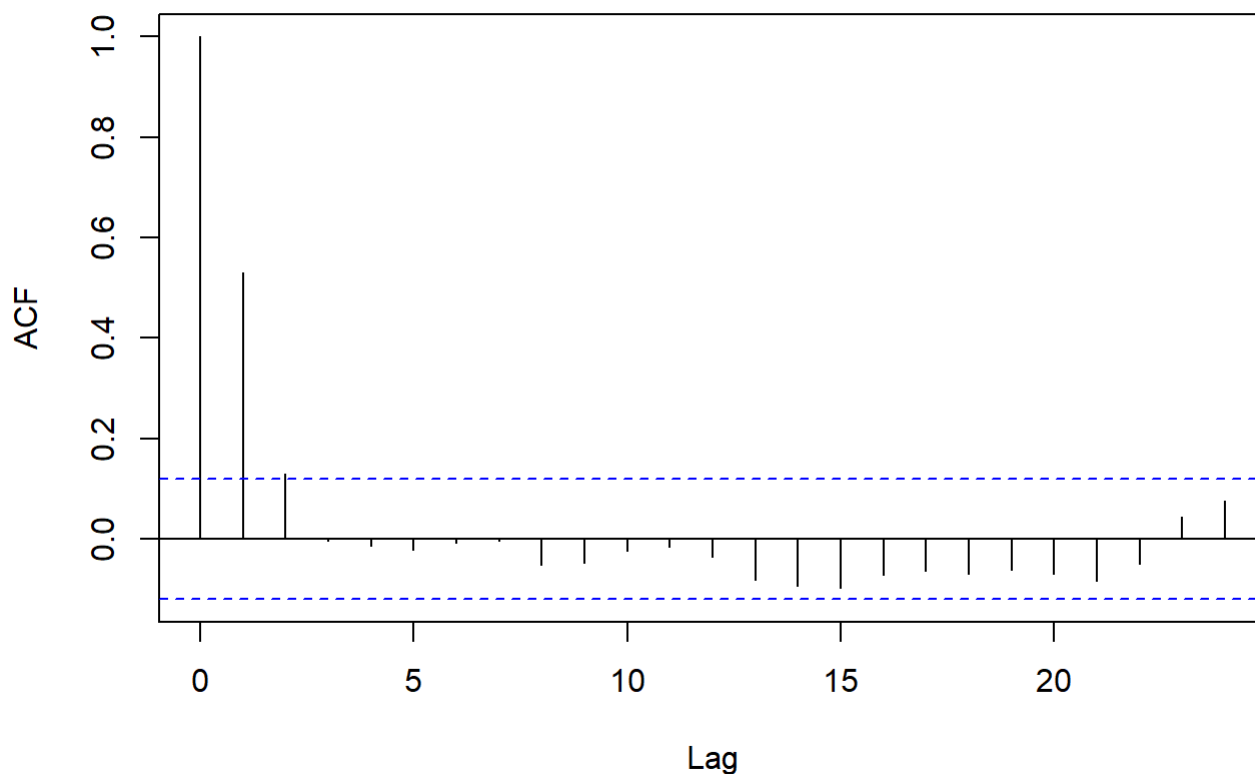
```
breakmodel3im = lm(labour_party~breakdBpl+breakpukip+breakpukipl+immigration+Refbreakim+Brexit  
acf(breakmodel3im$residuals)
```

Series breakmodel3im\$residuals



```
breakmodel4im = lm(liberal_democrats~breakdBpl+breakpukip+breakpukipl+immigration+Refbreakim+B  
acf(breakmodel4im$residuals)
```

Series breakmodel4im\$residuals



```
multivariatemodel = lm(cbind(reform_uk,conservative_party,labour_party,liberal_democrats)~brea
```

define a Seemingly Unrelated Regressions model structure

```
sur_model_imm = '
#Outcome Regressions with breakpoints#

reform_uk~breakdBpl+breakpukip+breakpukipl+ b1r*immigration + b2r*Refbreakim + b3r*Brexitbreak
conservative_party~breakdBpl+breakpukip+breakpukipl + b1c*immigration + b2c*Refbreakim + b3c*B
labour_party~breakdBpl+breakpukip+breakpukipl + b1l*immigration + b2l*Refbreakim + b3l*Brexitb
liberal_democrats~breakdBpl+breakpukip+breakpukipl + b1d*immigration + b2d*Refbreakim + b3d*Br
#The Covariance Structure of the Seemingly Unrelated Regressions#

reform_uk~~conservative_party+labour_party+liberal_democrats
conservative_party~~labour_party+liberal_democrats
labour_party~~liberal_democrats

#Contrasts#
diff_ref_con_pre := b1r-b1c
diff_ref_main_pre := (b1r-b1c)-(b1c-b1l)
diff_ref_con := (b2r+b3r+b4r)-(b2c+b3c+b4c) #How much larger are slopes after breaks between R
diff_con_lab := (b2c+b3c+b4c)-(b2l+b3l+b4l) # How much larger are slopes between Conservatives
diff_lab_lib := (b2l+b3l+b4l)-(b2d+b3d+b4d) # Between Labour and Lib Dem
diff_con_lib := (b2c+b3c+b4c)-(b2d+b3d+b4d) # Between Conservatives and Lib dems
diff_ref_main := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l))) # How different
```

```

PseudoDIDrefcon := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(b1r-b1c)
PseudoDIDrefmain := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l))) - ((b1r-b1c)
,

sur_model_inf = '
reform_uk~breakdBpl+breakpukip+breakpukipl + b1r*inflation + b2r*Refbreakinf + b3r*Brexitebreak
conservative_party~breakdBpl+breakpukip+breakpukipl + b1c*inflation + b2c*Refbreakinf + b3c*Brexitbreak
labour_party~breakdBpl+breakpukip+breakpukipl + b1l*inflation + b2l*Refbreakinf + b3l*Brexitebreak
liberal_democrats~breakdBpl+breakpukip+breakpukipl + b1d*inflation + b2d*Refbreakinf + b3d*Brexitebreak
#The Covariance Structure of the Seemingly Unrelated Regressions#

reform_uk~~conservative_party+labour_party+liberal_democrats
conservative_party~~labour_party+liberal_democrats
labour_party~~liberal_democrats

#Contrasts#
diff_ref_con_pre := b1r-b1c
diff_ref_main_pre := (b1r-b1c)-(b1c-b1l)
PseudoDIDrefcon := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(b1r-b1c)
PseudoDIDrefmain := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l))) - ((b1r-b1c)
diff_ref_con := (b2r+b3r+b4r)-(b2c+b3c+b4c) #How much larger are slopes after breaks between R
diff_con_lab := (b2c+b3c+b4c)-(b2l+b3l+b4l) # How much larger are slopes between Conservatives
diff_lab_lib := (b2l+b3l+b4l)-(b2d+b3d+b4d) # Between Labour and Lib Dem
diff_con_lib := (b2c+b3c+b4c)-(b2d+b3d+b4d) # Between Conservatives and Lib dems
diff_ref_main := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l))) # How different
,

```

fit a SUR

```

set.seed(123)
fitsurimm = sem(sur_model_imm,data=data_clean,se="bootstrap",ncpus=12,bootstrap=2000,parallel=
fitsurinfl = sem(sur_model_inf,data=data_clean,se="bootstrap",ncpus=12,bootstrap=2000,parallel=
modelsummary::datasummary(reform_uk+conservative_party+immigration+labour_party+liberal_democr

```

Descriptive Stats

	mean	sd	median	IQR
reform_uk	4.00	7.07	1.00	3.00
conservative_party	5.35	6.86	4.00	2.00
immigration	15.96	5.63	14.00	7.00
labour_party	8.55	11.31	6.00	5.25
liberal_democrats	2.63	3.45	2.00	1.00
inflation	6.74	3.36	5.00	2.00


```

modelsummary::modelsummary(list("Immigration interest"=fitsurimm,
"Inflation Interest"=fitsurinfl),coef_map=c(
  "diff_ref_con := (b2r+b3r+b4r)-(b2c+b3c+b4c)"="Refom vs Conservatives",
  "diff_con_lab := (b2c+b3c+b4c)-(b2l+b3l+b4l)"="Conservatives vs Labour",
  "diff_lab_lib := (b2l+b3l+b4l)-(b2d+b3d+b4d)"="Labour vs Libdems",
  "diff_con_lib := (b2c+b3c+b4c)-(b2d+b3d+b4d)"="Conservatives vs Libdems",
  "diff_ref_main := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l)))"= "Reform Con
"PseudoDIDrefcon := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(b1r-b1c)"="SlopeDiDrefcon=difference_in_slo
"PseudoDIDrefmain := ((b2r+b3r+b4r)-(b2c+b3c+b4c))-(((b2c+b3c+b4c)-(b2l+b3l+b4l)))-((b1r-b1c)-
"diff_ref_con_pre := b1r-b1c" = "Difference in slopes pre break",
"diff_ref_main_pre := (b1r-b1c)-(b1c-b1l)"= "Difference in difference Reform-Con - Con-Lab"
),title="How differently did the political shocks affect parties")

```

How differently did the political shocks affect parties

	Immigration interest	Inflation Interest
Refom vs Conservatives	0.945 (0.126)	0.693 (0.384)
Conservatives vs Labour	-0.330 (0.149)	0.179 (0.202)
Labour vs Libdems	0.044 (0.157)	-0.635 (0.288)
Conservatives vs Libdems	-0.286 (0.059)	-0.455 (0.194)
Reform Conservative Gap vs Conservative Labour Gap	1.275 (0.208)	0.513 (0.419)
SlopeDiDrefcon=difference_in_slopes(After-Before)	1.138 (0.189)	0.719 (0.661)
SlopeDDDrefmain=difference_in_a_difference_of_slopes(After-Before)	1.364 (0.262)	0.381 (0.707)
Difference in slopes pre break	-0.193 (0.081)	-0.027 (0.311)
Difference in difference Reform-Con - Con-Lab	-0.089	0.132

	Immigration interest	Inflation Interest
	(0.091)	(0.329)
Num.Obs.	268	268
AIC	6240.8	6313.7
BIC	6377.2	6450.1

```

modelsummary::modelsummary(list("Immigration interest"=fitsurimm,
"Inflation Interest"=fitsurinfl),coef_map=c( "reform_uk ~~ conservative_party"="cov(reform,con
"reform_uk ~~ labour_party"="cov(reform,labour)",
"reform_uk ~~ liberal_democrats"="cov(reform,libdems)",
"conservative_party ~~ labour_party"="cov(conservatives,labour)",
"conservative_party ~~ liberal_democrats"="cov(conservatives,libdems)",
"labour_party ~~ liberal_democrats"="cov(labour,libdems)",
"reform_uk ~~ reform_uk"="var(reform)",
"conservative_party ~~ conservative_party"="var(conservatives)",
"labour_party ~~ labour_party"="var(labour)",
"liberal_democrats ~~ liberal_democrats"="var(libdems)"
),title="residual covariances")

```

residual covariances

	Immigration interest	Inflation Interest
cov(reform,conservatives)	10.139 (2.771)	14.322 (3.937)
cov(reform,labour)	22.883 (9.491)	25.636 (9.210)
cov(reform,libdems)	8.413 (2.432)	9.928 (2.557)
cov(conservatives,labour)	54.509 (22.075)	57.104 (21.673)
cov(conservatives,libdems)	17.113 (5.272)	18.195 (5.258)
cov(labour,libdems)	25.586 (7.990)	26.228 (7.630)

	Immigration interest	Inflation Interest
Num.Obs.	268	268
AIC	6240.8	6313.7
BIC	6377.2	6450.1

APPENDIX:Full model summaries

```
summary(fitsurimm,ci=TRUE)
```

lavaan 0.7-2 ended normally after 148 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	38
Number of observations	268

Model Test User Model:

Test statistic	0.000
Degrees of freedom	0

Parameter Estimates:

Standard errors	Bootstrap
Number of requested bootstrap draws	2000
Number of successful bootstrap draws	2000

Regressions:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
reform_uk ~						
brekdbpl	-4.239	2.924	-1.450	0.147	-11.137	0.406
breakpkp	55.215	15.724	3.511	0.000	18.188	90.513
brekpkpl	-55.235	15.516	-3.560	0.000	-89.346	-18.347
immigrtn (b1r)	0.055	0.041	1.356	0.175	-0.019	0.144
Refbrekm (b2r)	0.319	0.196	1.624	0.104	0.043	0.810
Brxtbrkm (b3r)	-4.975	1.377	-3.612	0.000	-8.057	-1.770
Hstlbrkm (b4r)	5.135	1.365	3.763	0.000	1.875	8.165
conservative_party ~						
brekdbpl	5.720	4.120	1.388	0.165	0.547	17.556
breakpkp	33.530	14.037	2.389	0.017	0.383	47.749
brekpkpl	-29.425	13.414	-2.194	0.028	-43.421	1.366
immigrtn (b1c)	0.248	0.096	2.589	0.010	0.087	0.459
Refbrekm (b2c)	-0.380	0.350	-1.086	0.277	-1.418	0.005
Brxtbrkm (b3c)	-2.517	1.187	-2.119	0.034	-3.710	0.349
Hstlbrkm (b4c)	2.431	1.125	2.161	0.031	-0.148	3.547
labour_party ~						
brekdbpl	9.321	5.866	1.589	0.112	0.050	24.200
breakpkp	32.247	13.831	2.332	0.020	-3.646	46.506
brekpkpl	-30.790	12.824	-2.401	0.016	-43.474	1.267

immigrtn (b1l)	0.352	0.091	3.876	0.000	0.200	0.554
Refbrekm (b2l)	-0.653	0.513	-1.273	0.203	-2.006	0.128
Brxtbrkm (b3l)	-2.143	1.182	-1.814	0.070	-3.343	0.943
Hstlbrkm (b4l)	2.660	1.079	2.466	0.014	-0.058	3.688
liberal_democrats ~						
brekdBpl	2.076	1.569	1.323	0.186	-0.454	6.170
breakpkp	13.202	4.380	3.014	0.003	1.232	17.334
brekpkpl	-12.299	4.319	-2.847	0.004	-16.980	-0.773
immigrtn (b1d)	0.144	0.076	1.884	0.060	0.013	0.310
Refbrekm (b2d)	-0.130	0.116	-1.123	0.261	-0.472	0.012
Brxtbrkm (b3d)	-1.049	0.369	-2.840	0.005	-1.377	-0.048
Hstlbrkm (b4d)	0.998	0.354	2.818	0.005	0.041	1.346

Covariances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.reform_uk ~~						
.conservtv_prty	10.139	2.771	3.658	0.000	4.626	15.701
.labour_party	22.883	9.491	2.411	0.016	6.756	43.141
.liberal_dmcrt	8.413	2.432	3.459	0.001	3.822	13.318
.conservative_party ~~						
.labour_party	54.509	22.075	2.469	0.014	16.339	100.240
.liberal_dmcrt	17.113	5.272	3.246	0.001	7.066	27.295
.labour_party ~~						
.liberal_dmcrt	25.586	7.990	3.202	0.001	10.411	41.840

Variances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.reform_uk	28.890	10.160	2.843	0.004	10.816	49.910
.conservtv_prty	41.660	15.795	2.638	0.008	14.315	73.244
.labour_party	107.300	40.863	2.626	0.009	34.212	190.765
.liberal_dmcrt	11.009	3.007	3.661	0.000	5.236	17.192

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
diff_ref_cn_pr	-0.193	0.081	-2.378	0.017	-0.369	-0.044
diff_ref_mn_pr	-0.089	0.091	-0.975	0.330	-0.274	0.080
diff_ref_con	0.945	0.126	7.514	0.000	0.714	1.208
diff_con_lab	-0.330	0.149	-2.208	0.027	-0.713	-0.117
diff_lab_lib	0.044	0.157	0.277	0.781	-0.194	0.433
diff_con_lib	-0.286	0.059	-4.864	0.000	-0.413	-0.182
diff_ref_main	1.275	0.208	6.138	0.000	0.946	1.766
PseudoDIDrefcn	1.138	0.189	6.012	0.000	0.792	1.534
PseudoDIDrefmn	1.364	0.262	5.207	0.000	0.910	1.952

```
summary(fitsurinfl,ci=TRUE)
```

lavaan 0.7-2 ended normally after 49 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	38
Number of observations	268

Model Test User Model:

Test statistic	0.000
Degrees of freedom	0

Parameter Estimates:

Standard errors	Bootstrap
Number of requested bootstrap draws	2000
Number of successful bootstrap draws	2000

Regressions:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
reform_uk ~						
brekdBpl	3.274	2.478	1.321	0.186	-1.060	8.664
breakpkp	-3.421	2.265	-1.510	0.131	-8.182	0.577
brekpkpl	-0.248	2.036	-0.122	0.903	-4.043	3.997
inflatin (b1r)	0.600	0.284	2.112	0.035	0.110	1.228
Refbrknf (b2r)	-0.361	0.371	-0.971	0.331	-1.212	0.261
Brxtbrkn (b3r)	-0.624	0.445	-1.402	0.161	-1.477	0.306
Hstlbrkm (b4r)	0.687	0.107	6.429	0.000	0.503	0.917
conservative_party ~						
brekdBpl	9.580	3.838	2.496	0.013	2.769	18.061
breakpkp	-0.590	2.847	-0.207	0.836	-6.601	4.854
brekpkpl	0.343	1.075	0.319	0.750	-1.877	2.377
inflatin (b1c)	0.627	0.295	2.122	0.034	0.134	1.305
Refbrknf (b2c)	-1.634	0.724	-2.258	0.024	-3.420	-0.549
Brxtbrkn (b3c)	0.702	0.736	0.953	0.341	-0.497	2.502
Hstlbrkm (b4c)	-0.058	0.087	-0.667	0.505	-0.209	0.134
labour_party ~						
brekdBpl	4.148	6.344	0.654	0.513	-7.434	18.252
breakpkp	14.542	4.577	3.178	0.001	4.700	22.837
brekpkpl	-5.795	2.434	-2.381	0.017	-11.565	-2.302
inflatin (b1l)	0.785	0.286	2.748	0.006	0.306	1.435
Refbrknf (b2l)	0.084	1.101	0.076	0.939	-2.408	1.919
Brxtbrkn (b3l)	-1.778	1.086	-1.638	0.101	-3.529	0.850
Hstlbrkm (b4l)	0.525	0.194	2.708	0.007	0.255	1.009
liberal_democrats ~						
brekdBpl	3.582	1.556	2.303	0.021	0.819	6.776
breakpkp	0.766	1.351	0.567	0.571	-1.911	3.437
brekpkpl	-1.070	0.653	-1.639	0.101	-2.439	0.050
inflatin (b1d)	0.359	0.225	1.596	0.110	-0.037	0.852
Refbrknf (b2d)	-0.483	0.250	-1.934	0.053	-1.081	-0.099
Brxtbrkn (b3d)	-0.099	0.323	-0.306	0.760	-0.714	0.608
Hstlbrkm (b4d)	0.047	0.038	1.238	0.216	-0.012	0.135

Covariances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.reform_uk ~~						
.conservtv_prty	14.322	3.937	3.637	0.000	6.824	21.966
.labour_party	25.636	9.210	2.783	0.005	9.781	44.536
.liberal_dmcrt	9.928	2.557	3.883	0.000	5.015	15.098
.conservative_party ~~						

.labour_party	57.104	21.673	2.635	0.008	19.262	101.905
.liberal_dmcrtcs	18.195	5.258	3.460	0.001	8.161	28.474
.labour_party ~~						
.liberal_dmcrtcs	26.228	7.630	3.438	0.001	11.668	41.496

Variances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.reform_uk	36.407	12.912	2.820	0.005	14.132	63.713
.conservtv_prty	44.450	15.866	2.802	0.005	16.503	76.355
.labour_party	107.379	39.123	2.745	0.006	36.833	185.184
.liberal_dmcrtcs	11.378	2.973	3.827	0.000	5.686	17.582

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
diff_ref_cn_pr	-0.027	0.311	-0.086	0.932	-0.653	0.612
diff_ref_mn_pr	0.132	0.329	0.401	0.688	-0.540	0.786
PseudoDIDrefcn	0.719	0.661	1.089	0.276	-0.605	2.026
PseudoDIDrefmn	0.381	0.707	0.539	0.590	-0.993	1.812
diff_ref_con	0.693	0.384	1.805	0.071	-0.073	1.451
diff_con_lab	0.179	0.202	0.888	0.375	-0.225	0.580
diff_lab_lib	-0.635	0.288	-2.202	0.028	-1.245	-0.111
diff_con_lib	-0.455	0.194	-2.350	0.019	-0.851	-0.104
diff_ref_main	0.513	0.419	1.224	0.221	-0.303	1.355